HERDLINK

Understanding the simulation engine

A graduate study guide

Equations, parameters and intervention scenarios

Follow the daily calculation from movement records to compartment states. Work through a complete example, compare intervention schedules, and learn how assumptions and time aggregation shape the results.

30 September 2026

Model identity: `daily-contact-v1`

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1 A route through the material

HerdLink connects a dated movement ledger to a deterministic compartment model. This guide follows the calculation from input records to plotted results. You will learn how the daily equations work, how to interpret the controls in their stated units, and how to design and read an intervention comparison. Basic calculus, probability and matrix algebra provide the mathematical background; the guide introduces the compartment modelling concepts as they arise.

The calculation has three layers: **recorded movement**, **exposure pressure**, and **simulated state**. Recorded animal counts weight directed routes. These weights combine with source infectious shares to produce exposure pressure. Pressure then changes the compartments of each region through a daily recurrence, while regional population totals remain fixed.

The examples use a generic model, synthetic population units and the supplied historical movement ledger. Their purpose is to make the equations and intervention mechanisms concrete. Chapter 13 explains the model assumptions and the additional observations needed for an empirical application.

Read Chapters 2–6 first and reproduce the two-region calculation with a calculator. Then study Chapters 7–10 while using the comparison overlay. Chapters 11–14 connect the interface to the code, research interpretation and exercises. Chapter 15 supplies a compact reference sheet and source map.

Study session	Reading and task	What you should be able to explain
Foundations	Chapters 2–3; trace one input record	Which quantities are observed, constructed and assumed
Daily mechanics	Chapters 4–6; reproduce the worked example	Why the recurrence conserves population and why newly exposed units wait
Reading results	Chapters 7–8; inspect daily and weekly views	Why a stock, a flow and an attribution require different aggregation
Interventions	Chapters 9–10; inspect schedules and three columns	What each preset actually closes and what stays active
Synthesis	Chapters 11–15; complete the exercises	How mechanisms, assumptions and comparison design shape a result

Plan about four to six hours, including the exercises. At the end, explain an intervention result by connecting five pieces: the affected routes, the first affected day, the state used to compute pressure, the compartments that receive new entries, and the statistic displayed.

From equations to interpretation

Read the guide at three levels. First, follow the **implementation**: which data and equations the program uses. Second, derive the **mathematical consequences**, such as population conservation and residence times. Third, consider **empirical interpretation**: how model quantities could be related to population measurements and observations. Each level builds on the one before it.

2 What the engine represents

Regions and compartments

A node represents a region, with fixed population N. At the start of a day, its population is divided into susceptible S, exposed E, infectious I and recovered R. Exposed units are in a latent stage before infectiousness. Recovered units have left the infectious state and remain protected until immunity loss, where that mechanism is enabled.

$$S_{i,t} + E_{i,t} + I_{i,t} + R_{i,t} = N_i$$

All four quantities are real numbers. A value such as $I = 0.02$ is a small deterministic compartment amount. Fractional stocks allow the equations to describe smooth changes in a population. The engine applies the same arithmetic to each run, so identical complete inputs produce identical output. A stochastic model would add sampled events and a mechanism for chance extinction.

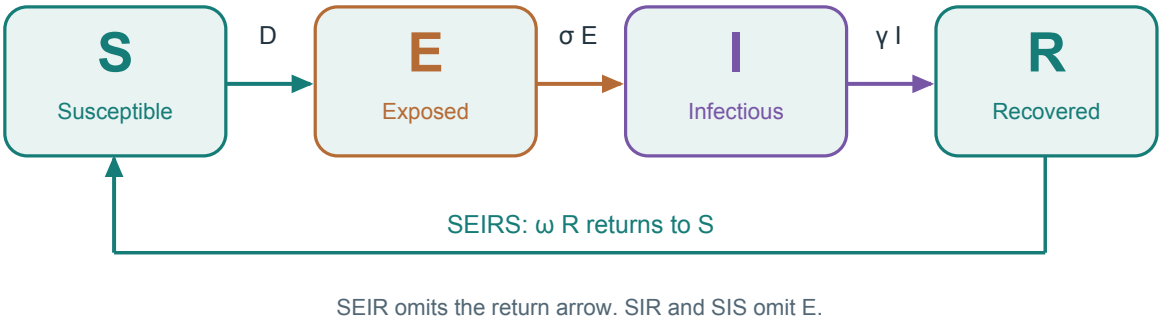


Figure 1: *Compartment transitions. Every arrow uses the state at the start of the same day.*

Two transmission pathways

Contact transmission is an assumed process within a region. Movement transmission depends on directed ledger records, including records whose source and destination are the same region. Their contributions add to the total hazard, so a local movement record and the contact term can both contribute on the same day.

Trade changes exposure pressure while each region retains its population N. The recurrence tracks changes in compartment membership within these fixed populations. This gives a compact representation of exposure through movement; demographic processes and physical relocation would require further state variables and balance equations.

Four model choices

Model	Compartment path	Distinctive assumption
SIR	S → I → R	No exposed stage; recovery removes susceptibility for the run
SIS	S → I → S	Recovery immediately returns units to susceptibility
SEIR	S → E → I → R	A latent stage precedes infectiousness
SEIRS	S → E → I → R → S	Recovered units can lose protection

In SIR and SIS, new infection entries join I at the end of the day. Their first contribution to transmission or recovery occurs in the following daily transition. In SEIR and SEIRS, new entries join E at the end of the day. Only the exposed units present at the day's start can progress to I during it.

Ledger mode and simulation mode

Ledger mode describes allowed recorded movements and network statistics. Simulation mode describes compartments and exposure calculations driven by the canonical daily ledger. The two modes answer complementary questions: Ledger mode describes the movement network, while Simulation mode describes the model response to that network. Metric selection changes the quantity displayed.

3 Inputs and population denominators

The daily ledger

Each usable record supplies a UTC calendar day, source region `COROP_LEV`, destination region `COROP_AFN`, and positive movement count `AANTAL`. The engine combines records sharing a day and directed route. $A \to B$ and $B \to A$ remain different routes. $A \to A$ is a recorded local movement.

The engine requires a consecutive daily observation calendar. An explicit calendar can include an empty movement day. The application constructs its calendar from recorded dates and checks for gaps, so each day needs a declared place in the input. Records with absent or `NA` endpoints are excluded; a known but undeclared endpoint raises an error. When preparing data, document both calendar coverage and the completeness of the recorded movements.

Synthetic population construction

The default denominator is a constructed scale of model units. In the available history before introduction, each region receives an activity total equal to outgoing volume plus 0.7 times incoming volume. A local record contributes to both parts. The history window is the 365 calendar days before introduction; records on or after introduction do not contribute.

$$T_i = V_i^{out} + 0.7V_i^{in}, \qquad z_i = \log(1 + T_i)$$

Let a and b be the smallest and largest positive z values. For a region with positive activity and $b > a$, the population construction is:

$$N_i = \text{round}(450 + 9550(z_i - a)/(b - a))$$

Zero activity receives 450 units. If the positive activities are all equal, every region receives 450 units. The 0.7 weight, logarithm and 450–10,000 range are modelling choices. Together they define the scale of the synthetic model population. The code stores this denominator in the field `holdings`; interpret its units through the population metadata.

A prepared inventory can instead provide a fixed regional vector with a declared reference, unit, date and geography. The inventory establishes the denominator and its units; transmission coefficients remain separate inputs. A region with $N = 0$ has undefined prevalence. Positive movement incident to such a region is rejected because the pressure calculation requires a positive population denominator, including when a scenario restricts that route.

Initialization

With share initialization, initial percentages apply to the seed region only. For $N = 10{,}000$, an initial infectious share of 1% supplies $I = 100$ in that region. The other regions begin susceptible. Exposed and recovered percentages are separate inputs and the three percentages must sum to at most 100.

Count initialization accepts explicit $S/E/I/R$ values for every region and requires initial percentages to be zero. This permits multiple initially infected regions. Each explicit vector must conserve its declared population. The tutorial script uses this form so that all starting values are visible.

At the exact introduction boundary the engine records an `initialFrame`, then computes the day's transition. Before introduction, compartments remain susceptible. Cumulative infection entries begin at zero, with initial E and I recorded as starting stocks. For a peak calculation, retain this initial boundary as well as every daily end state.

Changing the introduction date can change more than timing. With synthetic populations and share initialization it changes the historical population calculation; it also changes the historical graph used by several presets and the subsequent shipment calendar. For a timing comparison, record which of these inputs are held fixed.

4 The daily calculation

Step 1 Apply the day's permissions

Let $x_{ij,t}$ be the recorded count on a route that remains allowed after restrictions. A regional export restriction blocks routes from that region to other regions. A regional import restriction blocks routes from other regions into it. Neither closes its local route. A dated route restriction blocks the specified route for that calendar day, including a local route if explicitly selected.

All node changes dated on or before the day are applied before calculating pressure. Node permissions persist until another event changes them. Route restrictions are looked up for the current day only. The default permission is allowed.

Step 2 Compute movement pressure

The source infectious share determines how much pressure a permitted record supplies:

$$L_{ij,t} = \kappa x_{ij,t}(I_{i,t}/N_i)$$

κ is the Movement control. L has model population units for that day's exposure calculation. It measures the exposure contribution assigned to the route. All routes read the same start state, so exposure acquired today can influence later daily calculations.

Step 3 Combine the pathways at each recipient

The contact hazard is proportional to the recipient's own infectious share. Incoming movement pressure is divided by the recipient population:

$$h^c_{j,t} = \beta I_{j,t}/N_j$$

$$h^m_{j,t} = \sum_i L_{ij,t}/N_j, \qquad h_{j,t} = h^c_{j,t} + h^m_{j,t}$$

The movement sum includes $i = j$. For a fixed incoming pressure, a larger recipient denominator gives a smaller hazard. At low hazard and a fully susceptible destination, multiplying that hazard by the susceptible population approximately cancels this denominator in the incident count. The denominator still matters for fractions, depletion and subsequent dynamics.

Step 4 Convert hazard into new entries

$$p_{j,t} = 1 - \exp(-h_{j,t}), \qquad D_{j,t} = S_{j,t}p_{j,t}$$

D is the number of new infection entries during the day. In SEIR it enters E. The exponential keeps the entry fraction between zero and one for nonnegative finite hazard. At small hazard, $D \approx Sh$; at large hazard the susceptible stock limits entries.

The implementation uses `-Math.expm1(-h)` to evaluate $1 - \exp(-h)$ accurately when h is very small. The function evaluates the exponential difference accurately near zero.

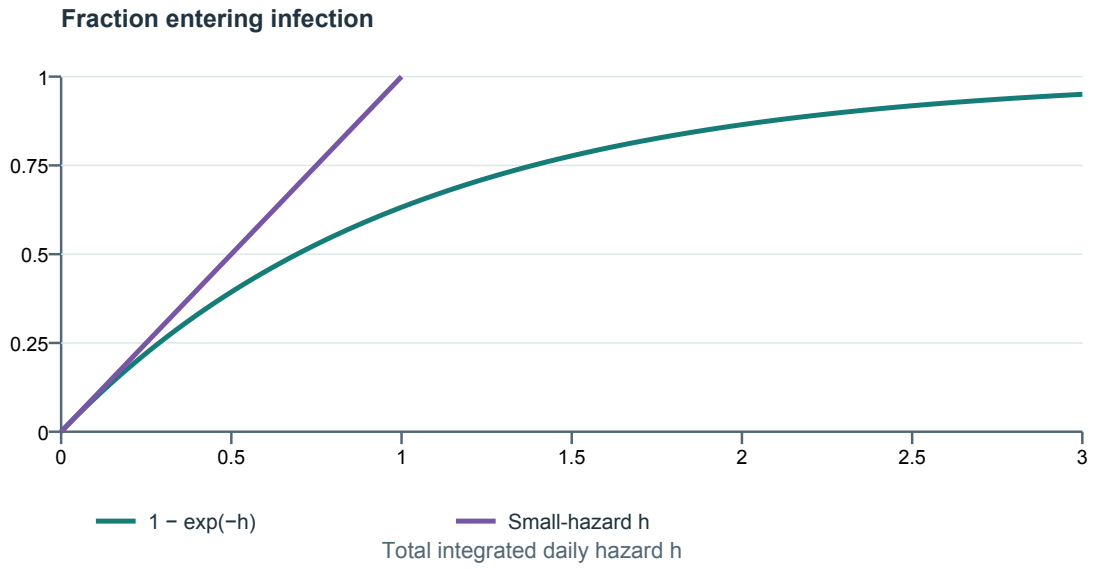


Figure 2: *The infection entry fraction and its linear approximation at small hazard.*

Step 5 Advance compartments simultaneously

Define progression $P = \sigma E$, recovery $G = \gamma I$, and immunity loss $W = \omega R$ for SEIRS. SEIR sets $W = 0$. Every quantity on the right uses the start state:

$$S_{t+1} = S_t - D_t + W_t, \qquad E_{t+1} = E_t + D_t - P_t$$

$$I_{t+1} = I_t + P_t - G_t, \qquad R_{t+1} = R_t + G_t - W_t$$

For SIR, $E = 0$, $P = D$ and $W = 0$. For SIS, $E = R = 0$, $P = D$ and $W = G$; the recovered amount goes directly back to S. To implement the update, calculate every flow from a copy of the starting compartments, then write all four end states together. This ordering gives each newly exposed unit its first opportunity to progress during the following day.

Step 6 Record states and flows

The engine records the end state, daily incident entries, cumulative entries, movement pressure and route attribution. A frame's `date` or `key` identifies the start of its interval; `intervalEnd` identifies when the stored compartment state is observed. This distinction becomes essential when reading weekly labels.

Why the recurrence conserves population

Adding the four updates cancels D, P, G and W. For valid controls, $0 \le D \le S$, $0 \le \sigma, \gamma, \omega \le 1$, and retained compartment terms are nonnegative. Thus conservation and nonnegativity follow directly from the equations. The engine checks these properties during calculation.

The model is a discrete daily recurrence: the equations above define one complete step. The exponential entry fraction treats the hazard as fixed within that step. Progression, recovery and immunity loss use daily fractions of the starting stocks. A continuous time model would instead describe simultaneous change within the day through coupled differential equations.

5 Understanding the controls

Control	Default	Meaning and units
Model	SEIR	Selects the compartment recurrence
Seed region	CR35	Region receiving the initial shares
Introduction	1 Jan 2019 for the bundled calendar	Exact initialization boundary
Initial infectious	1% of seed population	Starting infectious stock as a share of the seed population
Initial exposed	0%	Starting exposed share in SEIR or SEIRS
Initial recovered	0%	Starting recovered share outside SIS
Contact β	0.16	Daily integrated contact hazard at unit infectious share
Movement κ	0.04	Model units of pressure per recorded animal at unit source infectious share
Progression σ	0.22	Daily fraction of starting E entering I
Recovery γ	0.15	Daily fraction of starting I leaving infectiousness
Immunity loss ω	0.02	Daily fraction of starting R returning to S in SEIRS

Contact and movement coefficients are finite, nonnegative quantities and may exceed one. Progression, recovery and immunity loss are daily fractions in $[0, 1]$. Progression is active in SEIR and SEIRS; immunity loss is active in SEIRS. Choose the model first, then interpret the controls for the mechanisms it includes.

Residence time is geometric

An isolated compartment with daily exit fraction q retains a fraction $(1 - q)^k$ after k transitions. For $q > 0$, its mean occupied grid time is $1/q$ days. At the defaults, exposed and infectious mean grid times are about 4.55 and 6.67 days; recovered residence in SEIRS is 50 days. These means summarize a distribution of residence times: part of a cohort exits early and part remains for longer.

$$\Pr(T = n) = (1 - q)^{n-1} q, \qquad \mathrm{E}[T] = 1/q$$

At $q = 0$, an isolated cohort never exits. At $q = 1$, every member present at the day's start exits during that transition. Newly arriving members still wait until a later transition. For a continuous exit hazard r, matching one-day survival gives $q = 1 - \exp(-r)$. Matching a desired geometric mean D gives $q = 1/D$, when $D \geq 1$. Choose the conversion that matches the property being specified: survival over a day or mean residence on the daily grid.

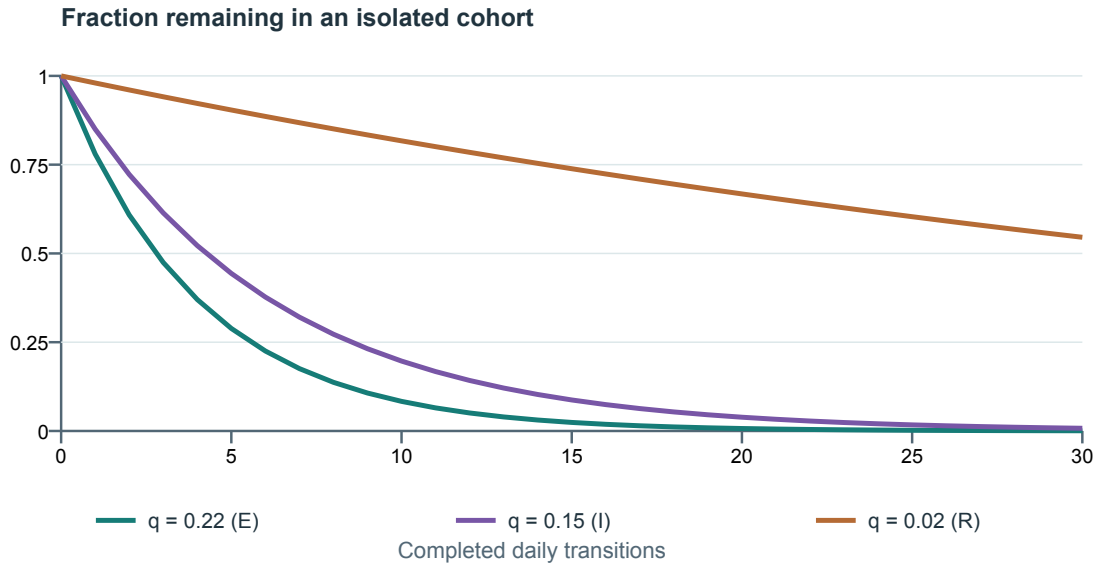


Figure 3: *Surviving fractions under three daily exit fractions. A long tail is part of the compartment assumption.*

What a parameter change can influence

Higher β increases the contact contribution at a fixed state. Higher κ increases movement pressure at a fixed state. Over a full trajectory, earlier growth can deplete susceptibles sooner and move the peak forward. Compare the peak, its date and cumulative entries to see these effects together.

Changing σ redistributes the timing of infectiousness. Changing γ changes both the duration of infectiousness and how much pressure can accumulate across days. Changing initial shares changes the starting stock and its timing relative to shipments. These mechanisms interact; a visually similar curve can arise from different combinations. Interpreting a fitted curve therefore also requires information about the quantities and processes used to estimate the parameters.

6 A complete two-region example

Consider fictional regions A and B, with populations 1,000 and 800 model units. Start with the following counts. There is one permitted record of 80 animals from A to B and no other movement. This SEIR example uses contact 0.10, movement 0.04, progression 0.22 and recovery 0.15.

Region	S	E	I	R	N
A	970	20	10	0	1,000
B	795	0	5	0	800

Pressure and incidence

A's infectious share is $10/1000 = 0.01$. Its export pressure is $0.04 \times 80 \times 0.01 = 0.032$. B receives movement hazard $0.032/800 = 0.00004$. B's contact hazard is $0.10 \times 5/800 = 0.000625$, making total hazard 0.000665.

B's new entries are $795 \times [1 - \exp(-0.000665)] = 0.528499$. A has no incoming movements and produces $970 \times [1 - \exp(-0.001)] = 0.969515$ contact entries.

A progresses $0.22 \times 20 = 4.4$ exposed units and recovers $0.15 \times 10 = 1.5$ infectious units. B has no exposed units to progress and recovers $0.15 \times 5 = 0.75$. The end state is:

Region	S at end	E at end	I at end	R at end	New entries
A	969.030485	16.569515	12.900000	1.500000	0.969515
B	794.471501	0.528499	4.250000	0.750000	0.528499

B's infectious count falls despite positive incidence, because incidence enters E and recovery removes existing I. A's infectious count rises because its pre-existing exposed stock progresses. Both populations remain exactly conserved up to floating-point precision. The movement record contributes pressure while A's compartment total remains 1,000.

Close A's exports at the start of this day

The A $\rightarrow$ B route disappears from the exposure calculation. A's own update is unchanged. B retains its contact hazard and produces $795 \times [1 - \exp(-0.000625)] = 0.496720$ entries. B still ends with $I = 4.25$ because its starting exposed stock and today's recovery flow are the same in both runs. The difference first appears in B's new exposed entries.

The one-day reduction in B's entries is about 0.031779. This is close to, but not exactly equal to, the 0.031789 entries attributed to the movement pathway in the open run. Attribution shares the total incident count among simultaneous hazards; removal of a pathway recomputes the nonlinear total. The two questions are distinct even in a single daily step.

Run the example

From the repository root:

```
node scripts/simulation-tutorial-example.mjs
```

The script imports the production engine, declares the fictional population, runs the open and closed cases, and checks the hand calculation, conservation and route attribution. It prints complete node states and the route fields. Read `initialStates`, `nodeInterventions`, `movementPressure`, `newInfections` and `riskLoad` together rather than inspecting just the final curve.

7 Reading pressure, incidence and prevalence

Movement pressure

L_{ij} measures the route's assumed exposure pressure before susceptible depletion and competing pathways. Regional incoming exposure and outgoing pressure sum these quantities over routes between different regions. They exclude local routes. Positive pressure can arrive at a region with no susceptible units and cause zero incident entries.

Infection entries and attribution

The destination's total incident entries are D_j. For a pathway with hazard h_k, the engine allocates:

$$D_{k,j} = D_j h_{k,j} / h_j \quad \text{when } h_j > 0$$

When total hazard is zero, all attributed entries are zero. The movement route receives the share corresponding to its own movement hazard. The local link collects contact attribution and

attribution from any local movement record. Therefore a local infection link can exist with zero recorded local trade: it represents the separate contact contribution.

Link `riskLoad` stores the attributed incident amount. Summing link attribution into a destination recovers its total incident entries. This accounting identity lets you connect each destination total to the hazards that contributed to its calculation.

Stocks, flows and cumulative counts

Prevalence is the infectious stock divided by population at an observation time. Incidence is the flow of new entries during an interval. Cumulative infections sum incident entries since initialization and exclude the introduced E/I stock. In SIS and SEIRS they count repeat entries, so they can exceed population. Their unit follows the population definition used for the run.

National prevalence is $\frac{\sum_i I_i}{\sum_i N_i}$, a population weighted mean of regional infectious shares. A large region contributes more weight. This distinction also explains why a striking regional peak can coexist with a small national percentage.

The pressure index

The internal field `rtProxy` divides outgoing pressure by infectious stock at the day's start. Over a displayed bin, it divides summed outgoing pressure by summed daily start-state infectious stock, recorded as `infectiousUnitDays`. It is undefined when the denominator is zero.

This measures outward movement pressure per infectious model unit per day. A reproduction number answers a different question: how many secondary infection entries arise over an infectious lifetime under specified conditions. Deriving it requires population denominators, transmission coefficients, compartment transitions and timing. Chapter 13 develops an example for a frozen movement matrix.

8 Time labels and plotted trajectories

Three clocks

There is a **ledger day**, a **model transition**, and a **display bin**. The first two are daily. The third may be daily, weekly, monthly or yearly. Each display resolution summarizes the same daily trajectory.

For a bin labelled 5 January covering [5January, 12January), the displayed I is the state after the 11 January transition, observed at the 12 January boundary. New infections are summed over the seven daily intervals. A closure on 8 January starts on 8 January inside this bin. Read the bin label as its start date and the stored stock as its endpoint value.

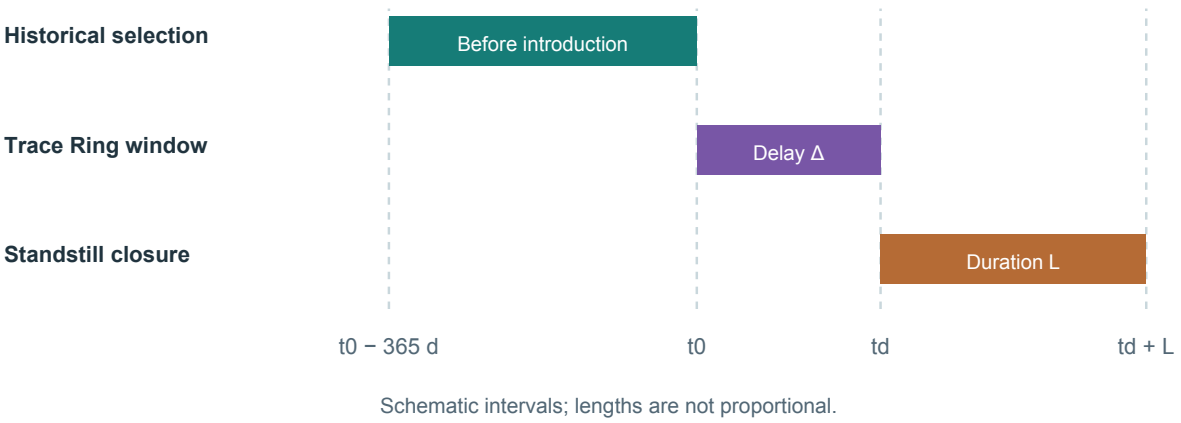


Figure 4: *Historical selection, tracing, response and reopening use distinct calendar intervals.*

Quantity	Display aggregation
S, E, I, R, prevalence	State at the actual end of the covered bin
New infections and attributed entries	Sum daily flows inside the bin
Incoming exposure and outgoing pressure	Sum daily pressure inside the bin
Cumulative infections	Cumulative value at the bin end
Population	The fixed vector or its total
Pressure index	Ratio of summed pressure to infectious unit-days

A partial final bin ends at the boundary of actual coverage. Summing a week's starting infectious stocks, each weighted by one day, produces infectious unit-days. Averaging daily prevalence gives mean prevalence over the week. The chart instead displays prevalence at the bin endpoint.

Exact peaks and timeline peaks

The three-column panel's Timeline peak is the maximum of its displayed trajectory. A weekly or monthly sample can miss the daily maximum. The headless evaluator measures its peak over the introduction state and daily end states. Use that daily path when reporting an exact maximum and its date.

Chart interpolation is a drawing convention between available points. The sampled values remain the observations used to draw the line. The three-column view shares scales between corresponding figures, so a visibly different height represents a numerical difference under that shared metric. A figure in another region row can use a different scale; the shared scale applies across the three columns for the same region.

Choosing a resolution for Ledger mode

Coarse Ledger mode evaluates aggregated records at their date labels. It can miss a closure inside a bin or apply start-date permissions to the whole aggregate. Simulation mode applies daily restrictions before computing its daily path. For exact retained-trade totals under dated closures, use the daily ledger or the headless evaluator.

Selecting a date in the comparison overlay inspects stored values. Editing a scenario setting changes its inputs and triggers a new calculation after the 600 ms pause.

9 The seven intervention scenarios

An intervention has four parts: a target rule, a route action, a start date, and a duration. A preset supplies those parts, while the simulation controls supply the compartment model and coefficients. Loading a preset in the single-scenario view replaces the existing intervention schedule. The three-column view instead builds and evaluates separate schedules without applying them to the live scenario.

Let introduction be t_0, response delay be Δ, and response be $t_d = t_0 + \Delta$. Dates are calendar days. Standstill lasts L days and applies on $[t_d, t_d + L)$: the opening boundary is included and the reopening boundary is excluded. A response affects the run from its start date through the remaining observation period.

Preset	Target rule	Action from response	What continues
Open trade	No targets	Clear restrictions for the reference	All movement and contact pathways
Seed containment	Seed region only	Close its exports to other regions	Imports into seed, local movement and contact
Trace Ring	Seed plus its direct outgoing recipients during tracing	Close each selected region's exports	Imports, local movement and contact
Community Cordon	Seed's historical trade community	Close routes leaving that community	Internal routes, incoming routes and contact
Hubs	Up to k historical outgoing-partner leaders	Close selected regions' exports	Other exports, imports, local movement and contact
Bridges	Up to k historical betweenness leaders	Close selected regions' exports	Other exports, imports, local movement and contact
Standstill	All regions	Close all routes between different regions for L days	Local movement and contact; cross-region trade after reopening

All persistent presets keep their target set fixed once constructed. The intervention acts through movement permissions. Contact transmission and compartment transitions continue under the selected simulation controls, and each allowed movement follows its recorded route.

Seed containment

This is an export action on the known seed. It removes later seed exports from the pressure calculation. Imports into the seed, local movements and contact transmission remain active. If recipients were already exposed before response, they retain those compartments and their own movement permissions. This is why a small delay can matter even when the seed itself is controlled for the rest of the run.

Trace Ring

The tracing window is $[t_0, t_d)$. The target set is the seed plus every direct recipient of a positive seed export in that interval. The window ends just before the response day. With zero delay, the window is empty and only the seed is targeted.

Selection uses one step of outward movement from the seed. Each direct recipient qualifies through a recorded export during the window, regardless of its simulated infection count. Incoming partners and further steps along the network lie outside this selection rule.

Longer response delays do two things together: they permit more days of movement before closure and allow a larger tracing target set. Consequently a Trace Ring delay comparison combines timing and scope. The two mechanisms can act in opposing directions, producing a nonmonotone sequence of outcomes.

Historical selection for Cordon, Hubs and Bridges

These strategies use positive routes between different regions in $[t_0 - 365\,\text{days}, t_0)$. Their application controls require a full preceding year. The graph and its targets are anchored to the history before introduction and remain fixed as response delay varies.

Repeated records on a route contribute to historical volume. Local records are excluded from this targeting graph but remain part of the simulation ledger and synthetic population calculation. Historical topology, historical population construction and displayed network statistics are three different uses of the data.

Hubs

For each eligible source, count its distinct historical outgoing destinations. Rank by that count, then outgoing volume, then region ID. Choose the first k, with at most the number of eligible regions. A source with many small destinations can outrank one with a much larger volume to a single destination. The ranking measures outward reach through distinct destinations.

Bridges

Construct a directed graph with an edge wherever historical volume is positive. Compute node betweenness from shortest directed paths measured in **hops**, giving equal shortest paths shared credit. Rank by that score, then outgoing partner count, then region ID. Only regions with at least one outgoing historical route are eligible; regions with zero betweenness remain eligible when they meet that condition.

If n_{ab} counts shortest directed paths from a to b and $n_{ab}(v)$ counts those passing through v, the score is:

$$B_v = \sum_{a,b} n_{ab}(v)/n_{ab}$$

The sum excludes pairs where v is an endpoint and excludes equal endpoints; an unreachable pair contributes zero. For the diamond A $\to$ B $\to$ D and A $\to$ C $\to$ D, B and C each receive one half of the credit for the pair A, D. Brandes' algorithm runs a breadth-first search from each source to count shortest paths, then traverses the search order backward to distribute dependency credit. It avoids enumerating every complete path.

The Bridges selector uses directed, unweighted Brandes betweenness. The Ledger panel uses weighted betweenness with inverse movement volume as distance. Read each metric with its path length definition, since the two definitions can produce different rankings. Brandes' publication supplies background on the algorithm; the exact graph, eligibility and tie rules are specified by HerdLink's selector code. Brandes publication record.

Community Cordon

For community selection, reciprocal historical volumes are combined into an undirected weight. Louvain modularity optimization supplies the partition. Broad uses resolution 1; Finer uses 1.5.

The groups represent trade connectivity. Louvain searches for a partition with high modularity; its result can be a local optimum. Geographic adjacency plays no role in that objective. A seed outside the historical graph forms a singleton group.

If C_s is the seed community, the closure rule is source $\in C_s$ and destination $\notin C_s$. Movement in the opposite direction remains open. The implementation applies this rule to recorded routes after response, including routes that were absent from the historical graph. Community membership stays fixed throughout the run.

The Louvain paper explains the general modularity method. HerdLink's historical window, symmetrization and resolution values come from its implementation. Blondel and colleagues. The displayed community panel describes a different graph and period, so its membership need not match the targeting partition.

Standstill

Standstill closes cross-region movement temporarily. Contact and recorded local movement remain active, and existing E/I continue through their compartment transitions. If those stocks survive until reopening, movement can spread pressure again. A short pause can change the time of a peak while changing the full-horizon cumulative count little. Closing more routes for a shorter duration and closing fewer routes persistently are different policy packages.

10 Comparing scenarios in the interface

Open Compare Scenarios with `C`, choose Simulation, and enable Three scenarios. Each column can use a strategy or a saved scenario. Preset columns use the active model settings and population; saved scenarios retain their own saved settings and population. Shared axes support visual comparison. To isolate an intervention effect, match the other inputs using each column's settings information.

Recommended set	Columns	Factor intended to vary
Strategies	Open trade, Seed containment, Trace Ring	Complete intervention strategy
Target counts	Hubs with 1, 3 and 5 targets	Number of selected regions
Response delays	Seed containment at 1, 7 and 14 days	Start of the same target action
Pause duration	Standstill for 7, 14 and 28 days	Time until movement reopens

Per-column inputs apply only where the strategy uses them. Target count controls Hubs and Bridges; the Trace Ring and Cordon selection rules determine their own group sizes. Pause duration controls Standstill. Open trade has no response action. Editing a column waits 600 ms before calculation; all three results are published together after they finish. The comparison columns maintain their own evaluation schedules alongside the live scenario.

The default region view shows three rows, selectable from one to five. It ranks regions by the peak of the selected metric in the active Original trajectory across the displayed timeline. All columns use these same reference regions, making corresponding rows easy to compare. Display sampling can influence the ranking because it determines the points available for the peak calculation. Saved scenarios also use this reference ranking.

A comparison worksheet

Before reading the curves, record the population reference and vector, model, coefficients, initial state, seed, introduction, canonical movement input and evaluation horizon. Then describe each schedule's exact targets, route directions, start and end. Changes to any of these inputs can contribute to a result difference.

For a simple timing lesson, set all columns to Seed containment with delays 2, 3 and 7 days. Keep all other inputs matched. Inspect both Prevalence and Cumulative infections. For a target-selection lesson, compare Hubs and Bridges at equal k, start date and duration, alongside Open trade. Compare retained movement as well: regions selected at equal target count can carry different amounts of trade.

Use a metric with a stated question. Peak infectious stock describes maximum burden; its date describes timing. Cumulative entries describe total model infection events over the horizon. Retained cross-region movements describe disruption to this recorded ledger. A strategy can improve one and worsen another. Interpret the trade-offs in relation to the question posed by the comparison.

For exact retained trade, sum permitted daily movement counts over the same post-introduction horizon and divide by the open reference's count. If that denominator is zero, the ratio is undefined. Retained trade measures the permitted share of the recorded ledger, with blocked traffic removed and the other route counts held fixed.

11 Understanding response timing

The bundled example introduces infection in CR35 on 1 January 2020. There are no recorded exports from CR35 to other regions on that day. On 2 January, it exports to ten regions. A one-day response starts at the beginning of 2 January and blocks those records. A two-day response starts on 3 January and allows the first export day. The first export day therefore creates a discrete change in which historical records contribute pressure.

The parameter regime determines what happens after escape. Under a hypothetical contact coefficient 0.32 and recovery fraction 0.15, contact alone can amplify a small introduction in a largely susceptible region. Once several regions have positive compartments, closing the seed's later exports can leave a similar eventual susceptible depletion. Peak timing may move even when final size changes little. Small deterministic fractional stocks can persist through repeated daily transitions.

The following example uses contact 0.10, movement 0.04, progression 0.22 and recovery 0.15. Contact alone is below its local growth threshold. Recorded movements, including local records, can still support growth. This setting illustrates how continued movement contributes to the trajectory.

Seed response delay	Daily network prevalence peak	Cumulative infection entries
1 day	0.0308%	641
2 days	0.0308%	697
3 days	0.0308%	739
7 days	0.0308%	834

These rounded values use the bundled daily ledger through 31 December 2023, the default synthetic population, CR35, introduction on 1 January 2020, initial infectious share 1%, SEIR, and persistent Seed containment. Peaks include daily observations; cumulative entries exclude

the introduced seed. Counts are expressed in the synthetic population units of the example. The daily peak is the same for these four schedules, while cumulative entries differ.

Response effects follow the shipment calendar and the evolving compartments. Weekend movement patterns, persistence of local transmission, susceptible depletion, target changes in Trace Ring, and Standstill reopening can all shape the result. Read both the size and timing of peaks, and use a common horizon long enough to include the features being compared.

12 From code to a reproducible result

The calculation chain

The application reads and validates controls, obtains a population snapshot, constructs dated restrictions, runs the shared daily engine, aggregates daily output, and projects selected metrics into charts. The unrestricted Original series reruns the same settings and daily input with empty intervention schedules. It supplies the model reference against which an intervention is compared.

- **Daily engine:** `src/runtime/simulation-engine.js`. Start with `validateSimulationSettings`, `prepareInputs`, `simulateDaily` and `aggregateDailyTrajectory` for validation, recurrence, accounting and aggregation.
- **Population:** `src/runtime/simulation-population.js`. Read `estimateSyntheticPopulation`, `createSyntheticPopulationSnapshot` and `validatePopulationSnapshot` for construction and metadata.
- **Target selectors:** `src/runtime/intervention-presets.js`. Read `getHistoricalPresetGraph`, the target selectors and `isCommunityCordonRoute` for historical graphs and selection rules.
- **Application runtime:** `src/runtime/herdlink-runtime.js`. Follow `buildPresetScenario`, `buildSimulationTrajectory`, `evaluateComparisonScenario` and `getOriginalSimulationSeries` from controls to calculated series.
- **Comparison interface:** `src/components/ScenarioComparison.jsx`. Follow the evaluation effect and metric projection for separate columns, delayed reruns and shared controls.
- **Charts:** `src/comparisonCharts.js`. Read `pairComparisonSeries`, `buildComparisonChart` and `topComparisonRegions` for alignment, scales and ranking.
- **Saved scenarios:** `src/scenarioStorage.js`. Read the validation and serialization routines for the saved record format.

The application caches daily trajectories by a canonical identity incorporating settings, population, dates, ledger and restriction schedules. Display sampling summarizes the daily model. A cache hit reuses a trajectory with the same complete input identity. Input validation checks the controls and scenario records before calculation.

Read a schedule as data

Node restrictions are a map from UTC midnight to a map of region permissions. For example, one event closes A's exports and a later event restores them. Each event changes its specified permission fields, so imports and exports can be controlled independently.

```
new Map([
  [Date.parse("2020-01-08"), new Map([["A", { exports: false }]])],
  [Date.parse("2020-01-22"), new Map([["A", { exports: true }]])],
])
```

This represents 14 restricted daily intervals. A route map keyed by 8 January and `A-B` represents a restriction on that route for that day only. The application expands a route edit made in a coarse display across the covered daily intervals. Read the resulting daily schedule to identify the exact restricted intervals.

Checking the calculation and assessing its use

Verification asks whether the program implements the declared equations. Validation asks whether the model supports a particular empirical use. The engine tests check fractional seeds, exact timing, state conservation, directed pressure, synchronous updates, attribution, undefined metrics and display invariance. These tests establish the numerical and software properties used throughout the worked examples. Empirical validation evaluates the relationship between model outputs and observations for a stated application.

Run the following from the repository root:

```
node scripts/simulation-tutorial-example.mjs
node --test tests/simulation-engine.test.js tests/intervention-presets.test.js
node --test tests/evaluation-harness.test.js
```

For the declared historical preset comparison, the evaluator imports the same engine and target selectors, constructs schedules without browser controls, and writes settings, metrics and input hashes:

```
node scripts/evaluate-scenario-presets.js reports/teaching-example --suite primary --seeds CR35
↪ --scales broad
```

The command writes the historical example as JSON and CSV, with separate fields for daily peaks, infection entries and permitted trade. Preserve the inputs and assumptions alongside any figure copied into a report.

What to retain with a saved result

Record engine identity, exact source version, coefficients, population metadata and values, initialization convention and state, introduction, daily data identity and coverage, restrictions, community scale, target rule, display resolution, metric definition and horizon. Saved scenarios use schema version 3 and `daily-contact-v1`. An inventory's source date and geographic definition establish the reference for its population counts.

When saved scenarios use different populations or model settings, interpret their comparison as a difference between complete input packages. For a comparison of interventions alone, match those inputs. Private inventories remain in browser memory; public scenario slots accept public population references.

13 Mathematical interpretation and model assumptions

Local growth near the susceptible state

Consider one region, no recorded movement, $S \approx N$, and small infectious amounts. Since $1 - \exp(-h) \approx h$, new entries are approximately βI. In SIR, the early infectious update is:

$$I_{t+1} \approx (1 - \gamma + \beta) I_t$$

Growth occurs when $\beta > \gamma$. In SEIR, the linearized updates are:

$$E_{t+1} \approx (1 - \sigma) E_t + \beta I_t, \qquad I_{t+1} \approx \sigma E_t + (1 - \gamma) I_t$$

For positive progression and recovery, this two-compartment system crosses its growth threshold at $\beta = \gamma$ as well. Progression affects the time scale of growth. This argument assumes near-complete susceptibility and removes all movement, including local recorded movement. With movement present, add its pressure contribution before interpreting growth.

From a movement matrix to a transmission matrix

For a frozen allowed movement matrix and positive population vector, linearization near susceptibility gives a transmission coefficient from infectious units in source i to new entries in destination j:

$$C_{ji} = \beta\delta_{ji} + \kappa x_{ij}/N_i$$

Here δ is one when source and destination agree and zero otherwise. Notice the source denominator after cancellation of the susceptible recipient population. For constant common recovery $\gamma > 0$, the expected number of occupied infectious grid intervals is $1/\gamma$. In this restricted frozen system, with positive SEIR progression and no loss from the exposed stage, a next-generation construction gives $K = C/\gamma$ and threshold $\rho(K) = 1$.

This derivation uses a frozen movement matrix and nearly complete susceptibility. In the daily model, shipments make the matrices time dependent, their chronological product governs propagation, and susceptible depletion changes the transmission process. The network panel describes the movement matrix itself; the construction above also incorporates transmission and recovery. General background on next-generation constructions is given by Diekmann, Heesterbeek and Roberts; the derivation above states its own restricted assumptions.

Population and coefficient interpretation

If every population and compartment amount is multiplied by a constant c while ledger counts stay fixed, scaling κ by c preserves prevalence trajectories under consistent initial shares. Contact coefficients and daily exit fractions stay unchanged. This transformation expresses the same prevalence dynamics in different model count units. Changing the relative population sizes of regions changes source shares and destination denominators by different factors, so it alters the model beyond a common unit conversion.

Relating model quantities to observations

The ledger observes dated movements. Connecting it to compartments requires further information about population sizes, infection states and transitions. Several combinations of coefficients, initial states and denominators can produce similar trajectories. Compartment counts, test positivity, reported cases and farm outbreak status each need their own relationship to the model state.

An empirical interpretation needs a stated unit and endpoint, population measurements at that unit, dated observations with denominators, an observation model, and evidence about reporting and diagnostic processes. Parameter estimation and uncertainty assessment then need a declared validation design. In a teaching exercise, settings can be chosen to expose a mechanism clearly. In an empirical study, the observation model and data determine how parameters are estimated and assessed.

The assumptions to carry into an analysis

The model assumes complete mixing within each region, fixed populations, constant coefficients, geometric exits, deterministic fractional states and one daily ordering convention. Movement interventions act on permissions while the contact coefficient remains fixed. Questions about

behaviour, rerouting, compliance, individual farms, shipment infection status or demographic turnover would require those mechanisms to be represented explicitly. Begin an analysis by deciding whether the stated assumptions fit the question being studied.

14 Exercises and worked answers

Attempt the questions before reading the answers. Use the fictional numbers to practise the equations and explain each result in words.

Questions

1. Units. A route carries 200 recorded animals, its source infectious share is 0.02, and $\kappa = 0.04$. The destination has population 1,000. Calculate route pressure and movement hazard. State what each number measures.

2. SEIR timing. A region starts with $E = 20$, $I = 10$, progression 0.22 and recovery 0.15. Today's new entries are 8. Calculate its end-state E and I. Identify which exposed stock supplies the progression flow.

3. Residence. With recovery 0.15 and no new arrivals into I, what fraction of an infectious cohort remains after seven days? What is its mean occupied grid time? Explain how these summaries describe a distribution of residence times.

4. Interval boundaries. A Standstill starts on 8 January and lasts 14 days. Is movement blocked on 21 January? On 22 January? How many restricted daily intervals are there?

5. Trace Ring. The seed sends to B before introduction, C on day 1 after introduction, D on day 2, and E on the day-3 response date. F sends into the seed on day 1. Which regions are targeted for a response delay of three days? Assume these are the only records.

6. Cordon. A community contains A and B. Which of A $\to$ B, A $\to$ C, C $\to$ A and A $\to$ A is blocked by its cordon? What happens to A's contact term?

7. Aggregation. A two-day bin has daily infection entries 2 and 3, infectious end states 10 and 12, and cumulative counts 7 and 10. What are the bin's new infections, infectious stock and cumulative infections?

8. Attribution. In the worked example, movement attribution is 0.031789 and the reduction after closing the route is 0.031779. Explain why these two values differ.

9. Population weighting. Region A has population 100 and infectious count 10; B has population 900 and infectious count 9. Calculate national prevalence and the unweighted regional average. Which does HerdLink display nationally?

10. Targeting. A has five outgoing historical partners and volume 100; B has two partners and volume 10,000. Which wins the Hubs ranking? Which additional outputs would you compare to assess model burden and movement disruption?

11. Local threshold. With no recorded movements, near-full susceptibility, $\beta = 0.10$ and $\gamma = 0.15$, calculate the SIR early multiplier. Explain why a region with those controls might still grow when movement is included.

12. Comparison design. Two saved scenarios differ in population, contact coefficient and target count. Their shared-scale plots differ. Explain what this comparison measures and describe a second comparison that isolates target count.

Worked answers

1. Pressure is $0.04 \times 200 \times 0.02 = 0.16$ model units for the day's calculation. Movement hazard is $0.16/1000 = 0.00016$. Pressure is the route contribution before division by destination population; hazard enters the exponential calculation. To obtain incident entries, combine all pathway hazards and multiply the entry fraction by the susceptible stock.

2. Progression is $0.22 \times 20 = 4.4$; recovery is $0.15 \times 10 = 1.5$. End-state $E = 20 + 8 - 4.4 = 23.6$; $I = 10 + 4.4 - 1.5 = 12.9$. Progression uses the 20 exposed units present at the start. The eight new entries become eligible to progress during the next transition.

3. The remaining fraction is $0.85^7 \approx 0.3206$, or 32.06%. Mean occupied grid time is $1/0.15 \approx 6.67$ days. Both describe a distribution. Some of the cohort exits early and some remains much longer.

4. Movement between different regions is blocked on 21 January and allowed on 22 January. The restricted interval is $[8\text{January}, 22\text{January})$, containing 14 daily steps. Local movement and contact continue throughout.

5. The target set is the seed, C and D. B is before introduction, E lies at the excluded response boundary, and F is an incoming-only partner. Selection ends with these direct recipients of the seed.

6. Only A $\rightarrow$ C is blocked. Internal, inward and local routes remain allowed. A's independent contact term remains active.

7. New infections are $2 + 3 = 5$; the infectious stock is 12; cumulative infections are 10. Flows sum across the bin; stocks and cumulative counts are read at its endpoint.

8. Attribution divides the incident count in one run among simultaneous hazards. Closing a route changes the total hazard and recomputes the exponential. The first value is a share within the open run; the second is the difference between two runs. Nonlinearity makes them slightly different.

9. National prevalence is $(10 + 9)/(100 + 900) = 1.9\%$. The unweighted mean of 10% and 1% is 5.5%. HerdLink displays the population-weighted national share, 1.9%.

10. A wins because Hubs first ranks distinct outgoing partners. Volume breaks only a tie in partner count. Compare infection entries or peak prevalence to assess model burden, and permitted movement totals to assess disruption to the ledger. An economic cost calculation would also need values assigned to that disruption.

11. The multiplier is $1 - 0.15 + 0.10 = 0.95$. Contact alone makes a small infectious stock decline in this approximation. Incoming pressure and recorded local movement add to this contact contribution and may change the direction of growth.

12. The first comparison measures the combined difference between the two complete scenario packages. To isolate target count, run both columns with the same population, coefficients, initial state, ledger, dates and target rule, varying only the number of selected regions. Shared scales then show the numerical difference for that matched comparison.

15 Reference sheet and sources

Essential equations

$$L_{ij} = \kappa x_{ij} I_i / N_i$$

$$h_j = \beta I_j / N_j + \sum_i L_{ij} / N_j, \qquad D_j = S_j[1 - \exp(-h_j)]$$

$$P_j = \sigma E_j, \qquad G_j = \gamma I_j, \qquad W_j = \omega R_j \text{ in SEIRS}$$

$$S' = S - D + W, \quad E' = E + D - P, \quad I' = I + P - G, \quad R' = R + G - W$$

Use start-state quantities on every right-hand side. For SEIR set $W = 0$; for SIR set $E = 0, P = D, W = 0$; for SIS set $E = R = 0, P = D, W = G$. A restriction acts through the allowed movement counts x.

Terms at a glance

Term	Meaning in this guide
Stock	An amount at an observation time, such as I
Flow	An amount accumulated over an interval, such as new entries
Hazard	The integrated exposure quantity entering the exponential
Pressure	Movement count × source infectious share × movement coefficient
Prevalence	Infectious stock divided by its population denominator
Counterfactual	A computed alternative under changed inputs within the model
Attribution	Allocation of one run's incident entries among assumed pathways
Verification	Checking implementation against its declared specification
Validation	Assessing fitness for a stated empirical use

Local sources

The equations and application behaviour in this guide follow the source tree inspected on 30 September 2026, using engine identity `daily-contact-v1`. The following files connect each topic to its implementation and tests.

- Daily engine and engine tests: recurrence, units, timing, accounting and aggregation.
- Population module and population notes: denominator construction and reference handling.
- Preset selectors and scenario specification: target rules, observation windows and actions.
- Application runtime and three-column component: control wiring, separate evaluations, timing and chart inputs.
- Model specification and scientific assessment: mathematical contract and scope of interpretation.
- Tutorial example: the fictional calculation in Chapter 6, with executable checks.
- Evaluation harness and harness test: historical example, reproducibility fields and checks on illustrative response separation.

Further reading

Brandes, U. (2001). A faster algorithm for betweenness centrality. *Journal of Mathematical Sociology*, 25(2), 163–177. DOI and publication record. Read for shortest-path dependency accumulation; HerdLink's exact targeting choices are documented in its source.

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., and Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics*, P10008. Author manuscript. Read for the modularity optimization underlying Louvain; Chapter 9 connects that method to HerdLink's historical window and cordon action.

Diekmann, O., Heesterbeek, J. A. P., and Roberts, M. G. (2010). The construction of next-generation matrices for compartmental epidemic models. *Journal of the Royal Society Interface*, 7(47), 873–885. Publication and abstract. Read for the distinction between a transmission model's next-generation construction and a raw adjacency matrix. Chapter 13 derives a frozen matrix example from HerdLink's recurrence.